

Lecture 1

Introduction to Electronic Communication

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Topics Covered (Chapter 1)

- Significance of Human Communication
- Communication Systems
- Types of Electronic Communication
- Modulation and Multiplexing
- The Electromagnetic Spectrum
- Bandwidth
- A Survey of Communication Applications

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1-1: Significance of Human Communication

- Communication is the process of exchanging information.
- Main barriers are language and distance.
- Contemporary society's emphasis is now the accumulation, packaging, and exchange of information.

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1-1: Significance of Human Communication

- Methods of communication:
 1. Face to face
 2. Signals
 3. Written word (letters)
 4. Electrical innovations:
 - Telegraph
 - Telephone
 - Radio
 - Television
 - Internet (computer)

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1-2: Communication Systems

- Basic components:
 - Transmitter
 - Channel or medium
 - Receiver
- **Noise** degrades or interferes with transmitted information.

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1-2: Communication Systems

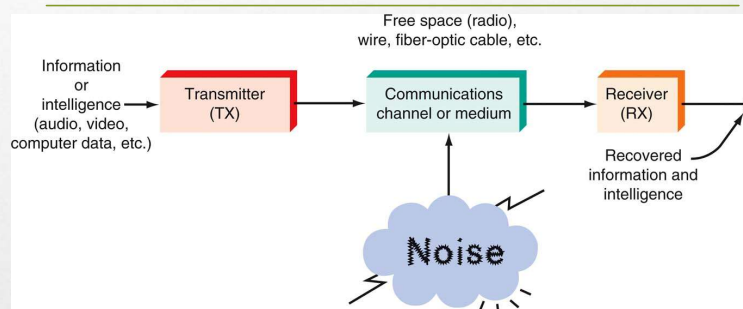


Figure 1-2: A general model of all communication systems.

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1-2: Communication Systems

Transmitter

- The **transmitter** is a collection of electronic components and circuits that converts the electrical signal into a signal suitable for transmission over a given medium.
- Transmitters are made up of oscillators, amplifiers, tuned circuits and filters, modulators, frequency mixers, frequency synthesizers, and other circuits.

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1-2: Communication Systems

Communication Channel (Transmission Medium)

- The **communication channel** is the medium by which the electronic signal is sent from one place to another.
- Types of media include
 - Electrical conductors (twisted pair, coaxial cable)
 - Optical media (fiber)
 - Free space (wireless)
 - System-specific media (e.g., water is the medium for sonar).

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1-2: Communication Systems

Receivers

- A **receiver** is a collection of electronic components and circuits that accepts the transmitted message from the channel and converts it back into a form understandable by humans.
- Receivers contain amplifiers, oscillators, mixers, tuned circuits and filters, and a **demodulator** or detector that recovers the original intelligence signal from the modulated carrier.

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1-2: Communication Systems

Transceivers

- A **transceiver** is an electronic unit that incorporates circuits that both send and receive signals.
- Examples are:
 - Telephones
 - Fax machines
 - Handheld CB radios
 - Cell phones
 - Computer modems

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1-2: Communication Systems

- Attenuation
 - Signal **attenuation**, or degradation, exists in all media of wireless transmission. It is proportional to the square of the distance between the transmitter and receiver.
- Noise
 - Noise is random, undesirable electronic energy that enters the communication system via the communicating medium and interferes with the transmitted message.

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1-3: Types of Electronic Communication

- Electronic communications are classified according to whether they are
 1. One-way (simplex) or two-way (full duplex or half duplex) transmissions
 2. Analog or digital signals.

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1-3: Types of Electronic Communication

Simplex

- The simplest method of electronic communication is referred to as **simplex**.
- This type of communication is one-way. Examples are:
 - Radio
 - TV broadcasting
 - Beeper (personal receiver)

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1-3: Types of Electronic Communication

Full Duplex

- Most electronic communication is two-way and is referred to as **duplex**.
- When people can talk and listen simultaneously, it is called **full duplex**. The telephone is an example of this type of communication.

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1-3: Types of Electronic Communication

Half Duplex

- The form of two-way communication in which only one party transmits at a time is known as **half duplex**. Examples are:
 - Police, military, etc. radio transmissions
 - Citizen band (CB)
 - Family radio
 - Amateur radio

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1-3: Types of Electronic Communication

Analog Signals

- An **analog signal** is a smoothly and continuously varying voltage or current. Examples are sine wave, voice, video (TV)

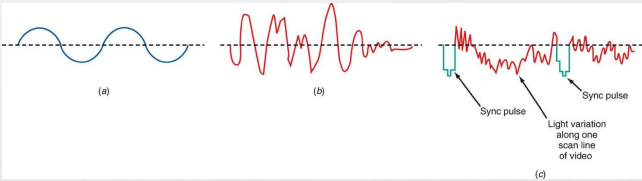


Figure 1-5: Analog signals (a) Sine wave “tone.” (b) Voice. (c) Video (TV) signal.

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1-3: Types of Electronic Communication

Digital Signals

- Digital signals change in steps or in discrete increments.
 - Most digital signals use binary or two-state codes.
- Examples are:
- Telegraph (Morse code)
 - Continuous wave (CW) code
 - Serial binary code (used in computers)

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1-3: Types of Electronic Communication

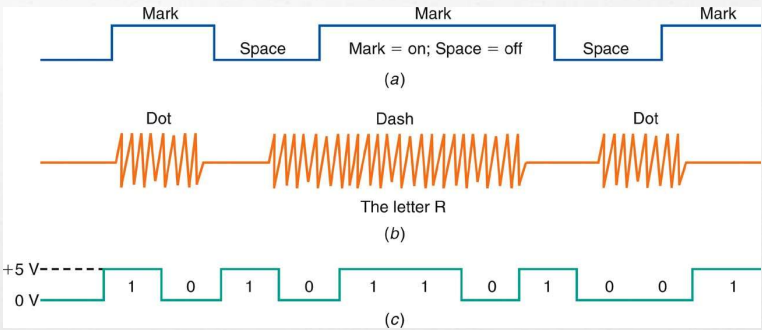


Figure 1-6: Digital signals (a) Telegraph (Morse code). (b) Continuous-wave (CW) code. (c) Serial binary code.

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1-3: Types of Electronic Communication

Digital Signals

- Many transmissions are of signals that originate in digital form but must be converted to analog form to match the transmission medium.
 - Digital data over the telephone network.
- Analog signals can be transmitted digitally.
 - They are first digitized with an analog-to-digital (A/D) converter.
 - The data can then be transmitted and processed by computers and other digital circuits.

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1-4: Modulation and Multiplexing

- **Modulation** and **multiplexing** are electronic techniques for transmitting information efficiently from one place to another.
- **Modulation** makes the information signal more compatible with the medium.
- **Multiplexing** allows more than one signal to be transmitted concurrently over a single medium.

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1-4: Modulation and Multiplexing

Baseband Transmission

- **Baseband** information can be sent directly and unmodified over the medium or can be used to modulate a carrier for transmission over the medium.
 - In telephone or intercom systems, the voice is placed on the wires and transmitted.
 - In some computer networks, the digital signals are applied directly to coaxial or twisted-pair cables for transmission.

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1-4: Modulation and Multiplexing

Broadband Transmission

- A **carrier** is a high frequency signal that is modulated by audio, video, or data.
- A **radio-frequency (RF) wave** is an electromagnetic signal that is able to travel long distances through space.

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1-4: Modulation and Multiplexing

Broadband Transmission

- A broadband transmission takes place when a carrier signal is modulated, amplified, and sent to the antenna for transmission.
- The two most common methods of modulation are:
 - Amplitude Modulation (AM)
 - Frequency Modulation (FM)
- Another method is called **phase modulation (PM)**, in which the phase angle of the sine wave is varied.

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1-4: Modulation and Multiplexing

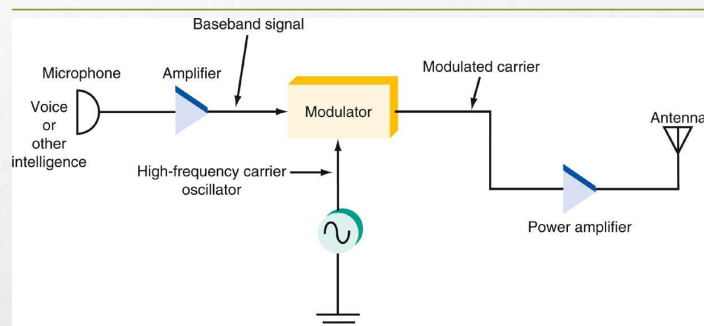


Figure 1-7: Modulation at the transmitter.

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1-4: Modulation and Multiplexing

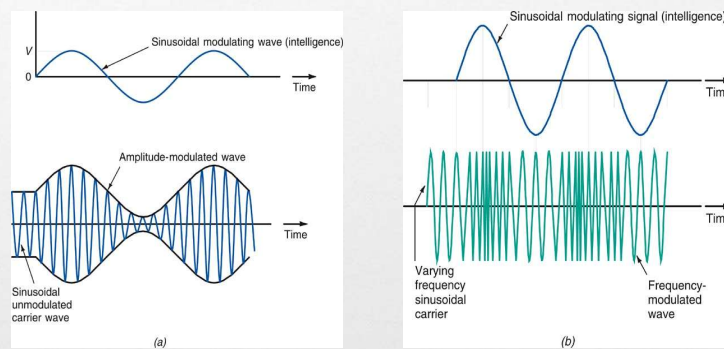


Figure 1-8: Types of modulation. (a) Amplitude modulation. (b) Frequency modulation.

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1-4: Modulation and Multiplexing

Broadband Transmission

- **Frequency-shift keying (FSK)** takes place when data is converted to frequency-varying tones.
- Devices called **modems** (**modulator-demodulator**) translate the data from digital to analog and back again.
- **Demodulation** or detection takes place in the receiver when the original baseband (e.g. audio) signal is extracted.

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1-4: Modulation and Multiplexing

Multiplexing

- Multiplexing is the process of allowing two or more signals to share the same medium or channel.
- The three basic types of multiplexing are:
 - Frequency division
 - Time division
 - Code division

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1-4: Modulation and Multiplexing

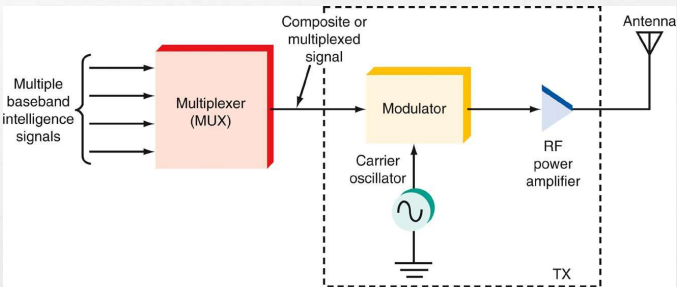


Figure 1-11: Multiplexing at the transmitter.

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1-5: The Electromagnetic Spectrum

- The range of electromagnetic signals encompassing all frequencies is referred to as the **electromagnetic spectrum**.

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1-5: The Electromagnetic Spectrum

The diagram illustrates the electromagnetic spectrum with two main scales: Wavelength (top) and Frequency (bottom). The Wavelength scale ranges from 10^7 m to 10^{-3} m. The Frequency scale ranges from 30 Hz to 300 GHz. The spectrum is divided into several regions: Radio waves (from 10^7 m to 10^{-3} m), Infrared (from 10^{-6} m to 10^{-7} m), Visible light (from 0.7×10^{-6} m to 0.4×10^{-6} m), Ultraviolet (from 0.4×10^{-6} m to 10^{-8} m), and X-rays, gamma rays, cosmic rays, etc. (from 10^{-8} m to 10^{-11} m). The optical spectrum (Infrared, Visible light, and Ultraviolet) is highlighted in green. The diagram also shows the relationship between wavelength and frequency, with a note that Wavelength decreases as Frequency increases.

Figure 1-13: The electromagnetic spectrum.

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1-5: The Electromagnetic Spectrum

Frequency and Wavelength: Frequency

- A signal is located on the frequency spectrum according to its frequency and wavelength.
- **Frequency** is the number of cycles of a repetitive wave that occur in a given period of time.
- A cycle consists of two voltage polarity reversals, current reversals, or electromagnetic field oscillations.
- Frequency is measured in cycles per second (cps).
- The unit of frequency is the hertz (Hz).

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1-5: The Electromagnetic Spectrum

Frequency and Wavelength: Wavelength

- **Wavelength** is the distance occupied by one cycle of a wave and is usually expressed in meters.
- Wavelength is also the distance traveled by an electromagnetic wave during the time of one cycle.
- The wavelength of a signal is represented by the Greek letter lambda (λ).

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1-5: The Electromagnetic Spectrum

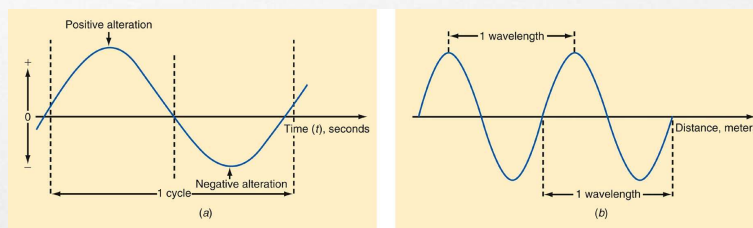


Figure 1-15: Frequency and wavelength. (a) One cycle. (b) One wavelength.

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1-5: The Electromagnetic Spectrum

Frequency and Wavelength: Wavelength

Wavelength (λ) = speed of light $\div$ frequency

Speed of light = 3×10^8 meters/second

Therefore: $\lambda = 3 \times 10^8 / f$

Example:

What is the wavelength if the frequency is 4MHz?

$\lambda = 3 \times 10^8 / 4 \text{ MHz} = 75 \text{ meters (m)}$

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1-5: The Electromagnetic Spectrum

Frequency Ranges from 30 Hz to 300 GHz

- The electromagnetic spectrum is divided into segments:

Extremely Low Frequencies (ELF)	30–300 Hz.
Voice Frequencies (VF)	300–3000 Hz.
Very Low Frequencies (VLF)	include the higher end of the human hearing range up to about 20 kHz.
Low Frequencies (LF)	30–300 kHz.
Medium Frequencies (MF)	300–3000 kHz, AM radio 535–1605 kHz.

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1-5: The Electromagnetic Spectrum

Frequency Ranges from 30 Hz to 300 GHz

High Frequencies (HF) (short waves; VOA, BBC broadcasts; government and military two-way communication; amateur radio, CB.	3–30 MHz
Very High Frequencies (VHF) FM radio broadcasting (88–108 MHz), television channels 2–13.	30–300 MHz
Ultra High Frequencies (UHF) TV channels 14–67, cellular phones, military communication.	300–3000 MHz

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1-5: The Electromagnetic Spectrum

Frequency Ranges from 30 Hz to 300 GHz

Microwaves and Super High Frequencies (SHF) Satellite communication, radar, wireless LANs, microwave ovens	1–30 GHz
Extremely High Frequencies (EHF) Satellite communication, computer data, radar	30–300 GHz

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1-5: The Electromagnetic Spectrum

Optical Spectrum

- The **optical spectrum** exists directly above the millimeter wave region.
- Three types of light waves are:
 - Infrared
 - Visible spectrum
 - Ultraviolet

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1-5: The Electromagnetic Spectrum

Optical Spectrum: Infrared

- Infrared radiation is produced by any physical equipment that generates heat, including our bodies.
- Infrared is used:
 - In astronomy, to detect stars and other physical bodies in the universe,
 - For guidance in weapons systems, where the heat radiated from airplanes or missiles can be detected and used to guide missiles to targets.
 - In most new TV remote-control units, where special coded signals are transmitted by an infrared LED to the TV receiver to change channels, set the volume, and perform other functions.
 - In some of the newer wireless LANs and all fiber-optic communication.

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1-5: The Electromagnetic Spectrum

Optical Spectrum: The Visible Spectrum

- Just above the infrared region is the **visible spectrum** we refer to as **light**.
- Red is low-frequency or long-wavelength light
- Violet is high-frequency or short-wavelength light.
- Light waves' very high frequency enables them to handle a tremendous amount of information (the bandwidth of the baseband signals can be very wide).

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1-5: The Electromagnetic Spectrum

Optical Spectrum: Ultraviolet

- Ultraviolet is not used for communication
- Its primary use is medical.

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1-6: Bandwidth

- **Bandwidth (BW)** is that portion of the electromagnetic spectrum occupied by a signal.
- **Channel bandwidth** refers to the range of frequencies required to transmit the desired information.

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1-6: Bandwidth

Spectrum Management and Standards

- **Spectrum management** is provided by government agencies set up to control spectrum use.
 - Federal Communications Commission (FCC) and National Telecommunications and Information Administration (NTIA) are two such agencies in the U.S.
- **Standards** are specifications and guidelines necessary to ensure compatibility between transmitting and receiving equipment.

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1-7: A Survey of Communications Applications

- Simplex
 - AM and FM broadcasting
 - Digital radio
 - TV broadcasting
 - Digital television (DTV)
 - Cable television
 - Facsimile
 - Wireless remote control
 - Paging services
 - Navigation and direction-finding services
 - Telemetry
 - Radio astronomy
 - Surveillance
 - Music services
 - Internet radio and video

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1-7: A Survey of Communications Applications

- Duplex
 - Telephones
 - Two-way radio
 - Radar
 - Sonar
 - Amateur radio
 - Citizens radio
 - Family Radio service
 - The Internet
 - Wide-area networks (WANs)
 - Metropolitan-area networks (MANs)
 - Local area networks (LANs)

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